Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building **verified software workflows**, Python services, CI releases, and technical documentation for risk-aware, medical-device-quality, Linux/Windows software.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python platform workflows** for 500+ users, mapping requirements to APIs, validation logic, technical docs, release notes, support runbooks, monitored jobs, and maintainable delivery for reviews and audits.
- Improved **software reliability and traceability** by tuning jobs, indexed queries, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** with documented production handoffs for release review records.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Python and JavaScript services** handling 10,000+ daily requests, implementing REST APIs, file ingestion, automated checks, documented fixes, sprint work, and production workflows with reproducible QA notes.
- Supported **verification-style testing workflows** by reproducing OCR, persistence, and execution defects, documenting root causes, regression cases, fixes, and review notes while cutting turnaround **60%** across validation logs.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Created **Python validation pipelines** across 100,000+ records, producing reproducible metrics, logs, reports, review artifacts, quality decisions, documentation, audit notes, and model-behavior analyses for traceable review evidence.
- Documented **algorithm behavior and edge cases** for researchers, turning scripts into repeatable workflows with clear handoffs, traceable outputs, reviewed experiments, references, and decisions for reproducible QA review.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform services and ETL workflows** using Python, SQL, AWS Glue, and S3, improving response time by **30%** across integrations, releases, reviews, and handoffs for service readiness reviews.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito across validation flows, raising CI reliability by **68%** and reducing release regressions safely across documented release checks.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **service-oriented application modules** with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting 2M+ synchronized records across workflows, data processes, integrations, dashboards, reviews, and releases with integration QA notes.
- Automated **Docker build and test flows** with Jenkins, cutting release time by **50%** while improving repeatable deployments, troubleshooting docs, handoffs, environments, and releases across controlled release checks.

Skills

Languages: Python, C# familiarity, Shell/Bash, SQL, Java, C++, JavaScript, TypeScript
Medical SW: IEC 62304 awareness, FDA 21 CFR 820.30 awareness, risk analysis, requirements, V&V, traceability
Architecture: UML familiarity, sequence diagrams, class diagrams, components, services, gRPC familiarity, REST APIs
Platforms: Linux, Windows, daemon/service concepts, virtual environments, networking, CANOpen CiA425 awareness
Tools: GitHub, Artifactory familiarity, SonarQube familiarity, CI/CD, Jenkins, Jira, Confluence, RQM familiarity, VSCode

Projects

HMIS Live | *MERN, TypeScript, ABDM APIs, Sentry*
- Built **healthcare workflow software** integrating patient, provider, claims, notifications, monitoring, and validation features with secure APIs, traceable flows, production-style reliability, and clinical operations support needs safely end-to-end.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a **containerized REST processing API** with AWS deployment, CI/CD releases, validation checks, and reproducible execution, achieving **87%** classification accuracy with diagnostics and reporting outputs consistently across inputs.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and database management through Java services and TypeScript interfaces for developers and analysts efficiently.